

Meghapranay Raparla

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EDUCATION

Amrita Vishwa Vidyapeetham

2024 – 2028

B.Tech in Computer Science and Engineering · CGPA: 8.34 / 10

Coimbatore, India

- Expected graduation: May 2028 · Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Object-Oriented Programming, Computer Networks, Machine Learning

EXPERIENCE

Software Engineer Intern — Nordic Tech Design

Apr 2026 – Jun 2026

Industrial bolt quality-inspection systems, telecentric-camera machine vision

Gurgaon, India (On-site)

- Sole developer of the core inspection software on a 5-person team; built a sub-pixel dimensional metrology engine measuring thread pitch, major/minor diameter, and hole-to-edge distance, validated to $<2\mu\text{m}$ repeatability across 40 runs — on par with commercial inspection systems
- Developed a golden-comparison surface-defect detector with image alignment to eliminate lighting-induced false alarms; on the validation part it detected a surface pit missed by the human operator
- Built an automated cross-drilled hole counter from video of a rotating bolt; deduplication logic correctly counted 4 holes despite each hole appearing multiple times per revolution

PROJECTS

MENTOR — AI Learning Platform

TypeScript · Node.js · MCP

github.com/nitrostacklh/command-global

- Built a 3-service backend exposing 23 API endpoints, with versioned JSON schema contracts between services so each could be deployed independently without breaking its clients
- Wrote and maintained a 182-test automated suite (all passing); used Claude Code to scaffold test cases and reviewed every generated case, cutting test-writing time roughly in half
- Implemented a diff engine that compares a student's committed code against their original design specification and flags divergence between stated intent and actual build
- Validated on a hosted MCP runtime with 16/16 live assertions passing and zero hardcoded credentials

IFDMS — Concurrent Real-Time Focus Monitoring

Python · asyncio · WebSockets · Flask

github.com/pranayr710/ifdms

- Architected a concurrent system: an asyncio WebSocket server on a background thread serves a Chrome extension and desktop HUD, while a synchronous main loop runs 10+ ML modules over thread-safe shared state with producer-consumer IPC
- Diagnosed and resolved race conditions across the async/sync thread boundary using logging and state-snapshot tracing
- Built an AI intent classifier, adaptive focus scorer, intervention engine, procrastination detector, and exam-mode domain blocker using OS-level hosts-file manipulation

MicroJava AST Interpreter

Java · JavaFX · Language Processing

github.com/pranayr710/microjava-interpreter

- Full compiler pipeline: lexer → parser → AST → SVG visualizer → step-by-step interpreter
- Built a breakpoint debugger exposing the symbol table and simulated memory state at each execution step

RESEARCH EXPERIENCE

Undergraduate Researcher — Amrita Vishwa Vidyapeetham

2025 – Present

- AI Security (Advisor: Dr. Harini N.) — hybrid C2PA + CNN-ensemble media authentication pipeline: 94.2% detection accuracy, F1 = 0.93 on 10,000+ adversarial deepfake samples (StyleGAN2, ProGAN, Stable Diffusion); cryptographic fast path reduced average latency ~49%
- Agricultural Computer Vision (Advisor: Dr. Amit Agarwal) — fine-tuned YOLOv8 on 5,000+ annotated field images for in-field strawberry disease detection: 91.7% mAP@0.5

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C

Backend & Systems: Node.js, Spring Boot, Flask, REST APIs, JWT/RBAC, MySQL, SQLite, Docker, Linux

Concurrent & Distributed: asyncio, WebSockets, multi-threading, thread-safe shared state, producer-consumer IPC, MCP protocol

AI-Assisted Development: Claude Code and Cursor for implementation and debugging; GitHub Copilot for inline completion; CodeRabbit for automated review on every pull request

AI / ML / CV: PyTorch, OpenCV, YOLOv8, EfficientNet, CNNs, ensemble learning, sub-pixel metrology, anomaly detection

Tools & Practices: Git, GitHub, VS Code, IntelliJ, Postman, Agile/Scrum, SDLC

Problem Solving: 156 LeetCode problems solved — graphs, dynamic programming, trees, recursion